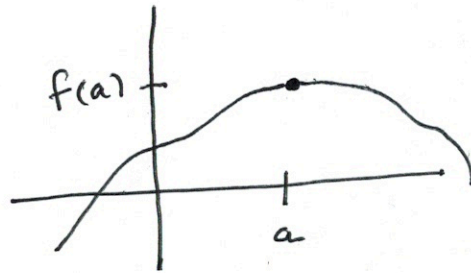


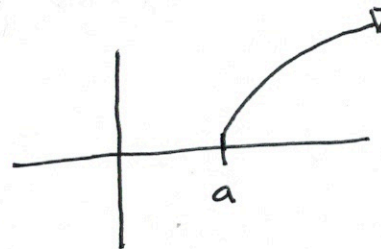
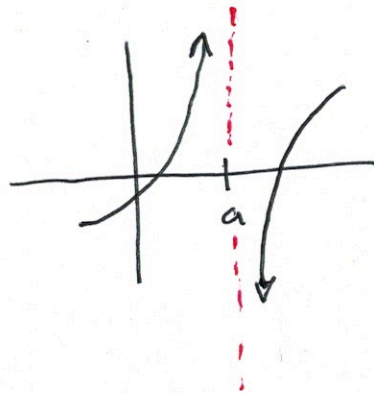
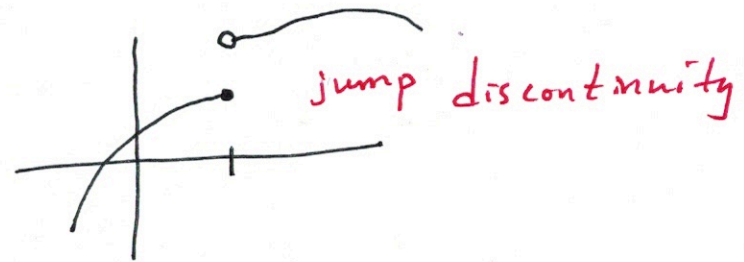
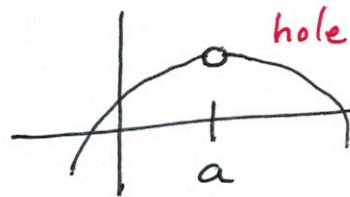
Section 2.5 Continuity

$f(x)$ is continuous at a



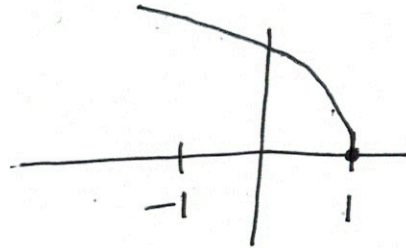
$$\lim_{x \rightarrow a} f(x) = f(a)$$

These are NOT continuous



Ex ① Is $g(x)$ continuous on $[-1, 1]$?

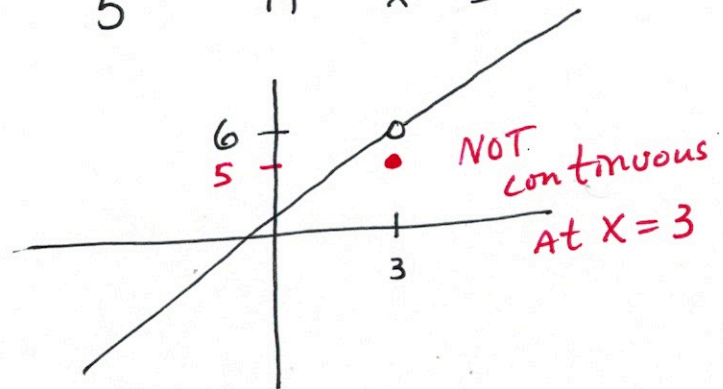
$$g(x) = \sqrt{1-x}$$



NOT continuous At $x=1$.

yes, it is continuous on $[-1, 1)$.

Ex ② $f(x) = \begin{cases} \frac{x^2-9}{x-3} & \text{if } x \neq 3 \\ 5 & \text{if } x = 3 \end{cases}$

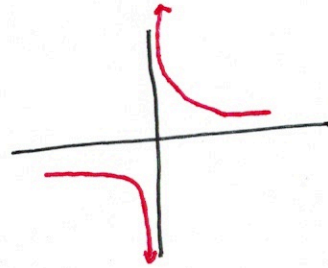


Ex (3) where is $f(x) = \sqrt[3]{x} (1+x^2)$ continuous.

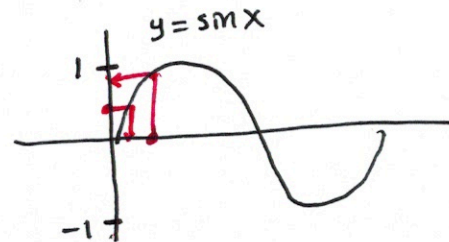
The domain of $f(x)$ is all real numbers,

So it is continuous everywhere.

$$g(x) = \frac{1+x^2}{\sqrt[3]{x}} \text{ NOT continuous at } x=0.$$



Ex (4) $f(x) = \sin^{-1}(x^2)$ where is it continuous?



the domain is $[-1, 1]$

Continuous on $(-1, 1)$

Ex (5) $f(x) = \frac{x^3 - 8}{x^2 - 4}$

what value of $f(2)$
will make $f(x)$ continuous

At $x = 2$.

$$\lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)}$$

$$= \frac{2^2 + 2(2) + 4}{2+2} = \frac{12}{4} = 3$$

So, make $f(2) = \underline{3}$.

